

UNIVERSITY OF CALIFORNIA

Santa Barbara

Connected by Tides: Missing Links in California's Coastal Ecotone

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy
in Ecology, Evolution, and Marine Biology

by

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December 2024

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Zoë Lucille Zilz

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PREVIEW

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PREVIEW

ABSTRACT

Connected by Tides: Missing Links in California's Coastal Ecotone

by

Zoë Lucille Zilz

On the boundary between land and sea, the Californian coastal ecotone is the location of many unique species interactions. My dissertation focuses on this unique and threatened strip of habitat and the trophic interactions therein. The objective of this body of work was to create new methodology and tools for deeper understanding of the coastal ecotone to promote conservation aims such as the preservation of habitat connectivity and ecosystem-based management. Chapter 1 outlines the construction of a comprehensive, highly resolved food web, inclusive of parasites, for the California rocky intertidal zone. This food web is the first of its kind for California and has the potential to provide managers, coastal conservation agencies, and marine protected area designers with high quality species interaction data, critical for understanding the cascading effects of species protection or loss. Included is a highly interactive online visualization tool that allows end users to subset and interrogate the food web network. Chapter 2 focuses in on one type of interaction within food webs, describing how our understanding of trophic networks changes when we incorporate parasites and their concomitant predation—or the accidental consumption of parasites by a predator—using the California rocky intertidal

food web as an example. I also compare the California rocky intertidal food webs to other similarly constructed high resolution food webs (e.g. inclusive of symbionts and concomitant predation) using commonly reported metrics such as trophic generality, connectance, and mean trophic level. Chapter 3 expands beyond the trophic interactions insular to the rocky intertidal zones and explores how terrestrial consumers interact with the coastal ecotone in remote and protected areas of Southern California. Specifically, I used camera traps to capture terrestrial wildlife moving throughout and foraging in ecotonal habitats (beaches, rocky reefs, estuaries, bluffs, and sand dunes) on California's remote Gaviota Coast. Camera trap footage revealed several terrestrial species consistently using these habitats, most notably coyote, deer, small rodents, and feral hogs. Mammal beach activity increased in times of terrestrial resource scarcity—when rainfall was scarce—and decreased in proximity to human disturbance. Terrestrial mammals, especially coyotes, were captured scavenging and hunting on beaches, dragging carcasses of marine prey (birds, marine mammals, invertebrates, and fish) from the intertidal to adjacent vegetated habitats. These previously unquantified interactions between terrestrial consumers and marine resources indicate a potential movement of marine to terrestrial resource subsidies, but only where there are no anthropogenic barriers to coastal access by wildlife. Much of the coastal ecotone is heavily exploited by humans for food, housing development, industry, and recreation especially in California. A comprehensive baseline understanding of how species interact in coastal ecotones is critical for ecosystem-based management in a time of global climate change, urban sprawl, and sea level rise.

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I. Introduction: Missing links in California's intertidal ecotone

The intertidal ecotone on the North American West Coast is an iconic ecosystem, not just because of its charisma and beauty, but also because studies of interactions between intertidal species have shaped many foundational ecological theories. The attraction to the intertidal is obvious; this stretch of coastline is one of the most biodiverse and productive in the world. Although the system is unique, ecological paradigms based on the structure and function of Eastern Pacific intertidal communities broadly apply. The keystone species hypothesis—that predation can control local diversity—was developed while Robert Paine threw the sea star *Pisaster ochraceus* into the sea and induced top-down changes in the sessile invertebrate community (Paine 1966, 1974). Joe Connell's work on barnacle communities gave rise to the concept of interspecific competition (Connell 1972). Paul Dayton laid the groundwork for the intermediate disturbance hypothesis in his monograph on the rocky intertidal zone (Dayton 1971). These nearly ubiquitous ecological rules help policymakers determine best practices and priorities in an age of science that has shifted its focus to management of a rapidly changing natural world. However, in an effort not to apply ecological theory to management in a reductionist manner, developing accurate and high-resolution baselines for communities will be paramount. Several important natural resources depend on intertidal habitat, either permanently residing there (red rock crabs, urchins), foraging there at low or high tide (seabirds, otter, sharks) or sheltering there as juveniles (opaleye, rockfish). Because the intertidal is a boundary ecosystem bordering marine and terrestrial habitats, protecting intertidal natural resources and ecosystem function is essential to effective, ecosystem-based management strategies. The northeastern Pacific coast has been the

subject of long-term ecological studies, including consistent monitoring by organizations like the Partnership for Interdisciplinary Studies of Coastal Oceans and the Multi-Agency Rocky Intertidal Network. Analyses of these long-term data have demonstrated large scale shifts in species distributions and community-wide response to climate change (Smith et al. 2006, Broitman et al. 2008, Blanchette et al. 2008). In short, we have rich, consistent, data on intertidal communities that is simultaneously spatially and temporally broad. Therefore, a comprehensive understanding of key species interactions within the intertidal ecotone has far reaching implications for conservation of nearshore ecosystems.

Ironically, one of the most cited papers discussing food webs focused on West Coast rocky intertidal organisms, but the “food web” discussed only included one predator and its prey (Paine 1966). This paper launched decades of research in the rocky intertidal zone. However, Paine’s seminal study is a perfect example of what’s missing from the massive body of research on the rocky intertidal – a comprehensive network of consumer-resource interactions. Despite significant research on interactions among some discrete groups of intertidal organisms, a well-resolved and comprehensive interaction network for the iconic Northeastern Pacific rocky intertidal zone ecosystem has yet to be assembled. My dissertation research fills in this critical gap.

In recent years, the value of resolving ecological networks has become evident, especially as selective defaunation by harvesting, habitat trampling, collecting, and climate change impacts ecosystems like the intertidal zone. Modern food webs networks are powerful analytical tools and enable robust statistical evaluations of ecological networks. Even basic food web analysis can facilitate refinement of ecosystem-based management. Information on network structure (properties of food web networks such as

stability, nestedness, connectedness) can be much more informative than metrics like species richness, which has previously been common focus of conservation efforts (Tylianakis et al. 2010). In intertidal areas that are difficult to access or data poor, a highly resolved food web would be an excellent tool for making management decisions without site-specific data. Food webs for some northeastern Pacific intertidal habitats, like the salt marsh estuaries, have already been assembled and used for conservation purposes such as assessing estuary restoration success.

Using interaction reports mined from the wealth of literature on the Eastern Pacific rocky intertidal zone, combined with data from my own collections, the Santa Barbara Channel LTER, PISCO, and MARiNe, I assembled a consumer-resource interaction matrix that formed the backbone for my dissertation. In subsequent chapters, I attempted to fill two remaining gaps in our understanding of California's intertidal zone communities: the importance of parasites and the role of terrestrial consumers. Parasites are important components of food webs in other marine ecosystems, and their inclusion in food webs promotes deeper understanding of trophic structure and ecosystem function. Terrestrial consumers are reported to consume marine resources worldwide, sometimes with cascading consequences for one or both ecosystems. Cumulatively, my dissertation research provides critical baseline data on two understudied communities in an otherwise well-studied ecosystem.

II. A parasite-inclusive food web for the California rocky intertidal zone

ABSTRACT

We present a highly resolved, species-rich food web, including parasitic interactions, for the California rocky intertidal zone. The food web is comprised of 1845 nodes, representing 1809 taxa, and 13,222 links representing trophic interactions between nodes. While only 670 links represent parasitic interactions, we have assembled possibly the most speciose parasite-inclusive food web ever published. The inclusion of all nodes and links are justified using multiple lines of evidence which are built into the dataset. In addition, metadata, including trophic strategy, taxonomic information, habitat, and other ecological attributes allow the data user to filter the food web to their specifications. The food web is a powerful and flexible tool for researchers with questions about large network properties, ecological dynamics of rocky shores, and the role of parasites in ecosystems. Our food web can be used to predict how complex ecosystems like the California rocky intertidal will respond to anthropogenic change and management strategies.

A. INTRODUCTION

The California rocky intertidal zone is an iconic habitat with a rich history of foundational ecology and natural history. Due to the biodiversity and ease of access of the rocky intertidal zone (RIZ) ecosystem, it became a hub of experimental ecology in the 1960s and 70s (Paine 1966, Dayton 1975); as a result, there is a comprehensive body of literature on RIZ interspecies interactions. Unfortunately, the accessibility and natural

appeal that facilitated decades of intertidal research also poses a conservation challenge for the RIZ, especially on California's heavily populated and developed coastline (Halpern et al. 2009). Human recreation, visitation, and harvesting erodes the very biodiversity that draws visitors to the RIZ, with variable consequences for the RIZ ecosystem, adjacent beaches, and subtidal reefs (Keough et al. 1993, Addessi 1994, Murray et al. 1999, Huff 2011, Bednar and Trulio 2017, Jhaveri and Smith 2019). Rocky intertidal habitats face multiple stressors in addition to direct anthropogenic disturbance, including sea surface temperature increase (Barry et al. 1995, Helmuth et al. 2006, 2011), sea level rise (Kaplanis et al. 2020), ocean acidification (Gaylord et al. 2011), storm intensification (Byrnes et al. 2011), introduced species (Culver and Kuris 2000, Zabin et al. 2018), and disease (Lafferty and Kuris 1993, Miner et al. 2006, Eisenlord et al. 2016, Harvell et al. 2019, Groner et al. 2021). Contemporary research on the RIZ has demonstrated large scale shifts in species distributions and community-wide response to climate change (Smith et al. 2006, Broitman et al. 2008, Blanchette et al. 2008). There is growing evidence that despite apparent long-term stability, the Northeastern Pacific RIZ is on the cusp of an ecological tipping point (Miner et al. 2021, Menge et al. 2022). Especially in extremely biodiverse ecosystems like the RIZ, species losses or shifts in abundance can be cryptic, resulting in subtle effects on ecosystem dynamics that go unnoticed until reversal is too late (Jurgens et al. 2015).

Scientists can use food webs as a tool to predict and mitigate the effects of anthropogenic disturbance. Food webs are valuable tools for contextualizing species interactions, modeling complex community structure, and reconciling data on pairwise relationships with larger-scale ecological patterns (Thompson et al. 2012). Food webs can

be useful to predict ecosystem response to perturbation (Dunne et al. 2016, Whitehouse and Aydin 2020, Lewis et al. 2022), ecosystem impacts of species loss (Petchey et al. 2008, Dunne and Williams 2009, Coll et al. 2011, Keyes et al. 2021), and can identify key conservation targets (Jordán 2009, MÉRILLET et al. 2022). Construction of highly resolved food webs, while challenging, can yield powerful tools for conservation, reflecting the true complexity of ecosystems. Ecological network structure (properties of food web networks such as stability, nestedness, centrality, and connectedness) can add depth to simple metrics such as species richness, which has previously been a common focus of conservation efforts (Tylianakis et al. 2010). Food webs represent a tangible link between basic research and conservation priorities. For example, loop analysis—an analytical tool demonstrating the effect of a sustained change in abundance of one species on the rest of the food web—can provide critical information to managers as they consider the downstream effects of either protecting or eradicating species (Carey et al. 2014). Furthermore, simulating press perturbations using conceptual food webs can help coastal managers identify conservation priorities (e.g. umbrella species) and strategies (Giakoumi et al. 2015). In the face of the Anthropocene, developing reliable high-resolution baselines for ecosystems is paramount. In RIZ areas that are difficult to access or are data poor, a highly resolved food web can be a critical tool for making management decisions without site-specific data.

Analytical representations of ecosystems must include their infectious agents. Parasitic organisms form the backbone of essential ecosystem properties and function. These consumers are abundant in other coastal food webs in California (Lafferty et al. 2006b, Hechinger et al. 2011, Morton et al. 2021) and in intertidal habitats worldwide